Project Development Phase — Dashboard Design

Date **Team ID**	_____ _____
Project Name	Visualization Tool for Electric Vehicle Charge and Range Analysis

1. Dashboard Overview

The EV Charge and Range Visualization Dashboard is a single-page web application. It presents seven interactive chart views accessible from a persistent left-side navigation panel. A filter bar at the top of every view allows users to refine data across all charts simultaneously. The dashboard is designed to require no training to use — a first-time visitor should be able to find and apply filters and interpret charts within two minutes.

2. Page Layout Structure

Zone	Description
Top Navigation Bar	Project title, Tableau Story link, Export button, Share button. Fixed at top; always visible on scroll.
Filter Panel (Top)	Horizontal row of filter controls: Model multi-select, Season toggle, Temperature slider, Terrain dropdown, Date range picker. Filters are persistent across all views.
Summary Stats Panel	Four large KPI cards below the filter bar: Total Sessions, Average Range (km), Average Charge Time (min), Most Efficient Model. Updates on every filter change.
Main Chart Area	The primary content area. Displays the currently selected chart view. Charts fill the available width. Tooltips appear on hover. Annotations are displayed directly on the chart.
Chart Navigation (Left) **Footer**	Vertical list of all seven chart views. Clicking any item loads that chart in the main area. Active chart is highlighted. Data source attribution, project name, team ID, and last dataset update date.

3. Chart Views — Detailed Specification

View	Chart Name	Chart Type	Axes / Dimensions	Key User Insight Delivered
1	**Overview / Home**	Summary KPI cards + mini sparklines	N/A — cards show totals and averages	Instant snapshot of dataset size and key averages before exploring deeper
2	**Range Heatmap**	Geographic heatmap (D3 + Leaflet)	Map geography; colour = energy	Where and on what route types does battery range deplete fastest

			consumption intensity	
3	**Charge Curve**	Multi-line chart (D3.js)	X: time (minutes); Y: battery percentage; line per model	How charging speed slows above 80%; real charge time vs. advertised
4	**Model Comparison**	Grouped bar chart (Plotly)	X: EV model; Y: real range (km); average line overlay	Which models deliver best real-world range vs. their spec sheet
5	**Temp vs. Range**	Scatter chart (Plotly)	X: temperature (°C); Y: range (km); colour = season	Quantified range drop in cold weather; seasonal range variation
6	**Degradation Trend**	Line chart (D3.js)	X: charge cycle count; Y: range (km); line per model	Which models retain range best over 100, 200, 500 charge cycles
7	**Export / Share**	Modal overlay panel (React)	N/A — export format selector and URL copy button	Download any chart or full dashboard; share with filters preserved

4. Filter Specification

Filter	UI Component	Options / Range	Charts Affected
Vehicle Model	Multi-select dropdown	15 EV model options + "All Models" default	All 7 charts
Season	Toggle buttons	Spring, Summer, Autumn, Winter, All	All 7 charts
Temperature Range	Dual-handle range slider	-30°C to +50°C; default: full range	All 7 charts
Terrain Type	Dropdown	All, Flat, City, Hilly, Highway	All 7 charts
Date Range	Calendar date picker (from/to)	Any date range within dataset period	All 7 charts
Charger Type	Dropdown	All, Home Slow, Public AC, Public DC Fast	Charge Curve, Comparison, Stats only